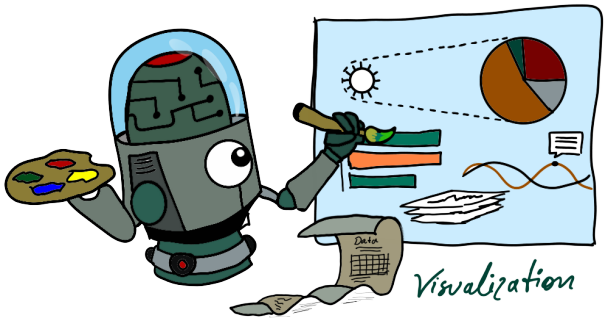
DS5343 - Visualización de Datos

UTEC
UNIVERSIDAD DE INGENIERÍA Y TECNOLOGÍA

University of Engineering and Technology



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Assignment: UtecVis: Network Visualization

Goal:

Design a single network visualization that uncovers key patterns of collaboration among University of Engineering and Technology - UTEC. You should focus on clear encoding of network metrics and leverage available visualization. Accompany your visualization with animation resource using visualization mantra.

Data:

In this activity, you will work with data on UTEC faculties members, including their **ProfessorID**, **Name**, **Department**, and **ResearchAreas**. You can obtain this information from the UTEC Collaborators website: Utec Collaborators website. Feel free to enrich your dataset with additional metrics—such as Google Scholar profiles, h-index scores, or other relevant indicators—to enhance your analysis.

Assignment

The task is to download the data using web-scrapping and create a network visualization that clearly communicates the relationships among the faculties members. Start by formulating a specific tasks (minimum 4 tasks) that your visualization will answer. **Design your chart to address that tasks**. Example tasks include:

- Who are the most prolific collaborators (highest degree or weighted degree)?
- Which professors act as bridges between departments (highest betweenness)?
- Which new collaborations are most likely to emerge among the faculty? (node prediction)
- ...

While you must use the provided dataset, you are free to filter, transform, or augment it as needed. You may also incorporate external data sources if it enhances your analysis. Ensure that your visualization stands on its own—readers should understand the insights.

Tip: Consider using natural language processing to help identify and structure the network relationships. You could inspire your visualization in the Figure **??** (use professors' photos as part of the nodes.).



Figura 1: Inspiration

Evaluation Criteria

This assignment is worth 20 points. I will evaluate both the strength of your design and the quality of your computing process. I will also consider factors such as the target audience, the clarity of your message, and how well the intended task is addressed. Your work will be grade based on:

- **Clarity**: Is the visualization easy to interpret?
- **Analytical Insight**: Does it reveal top collaborators and bridges?
- **Design Rationale**: Are your choices well-motivated?
- **Technical Accuracy**: Are network measures calculated correctly?
- **Tasks**: The visualization help to answer 4 tasks effectively.
- **Interaction:** It is mandatory to use interaction and animation resources such as: mouseenter, mouseover, mouseclick, and so on.

Deliverables instructions

- Your coded solutions must use D3. Other visualization libraries are not allowed.
- Submit to canvas a single ZIP containing a flask application and data files (nodes and edges).
- You are required to present your work in class. Failure to present will result in no grade for this assignment.